CH3

Q. What is data reduction? Explain different techniques of data reduction?

Data reduction:

⇒ Data reduction is a process that reduces the volume of original data and represents it in a much smaller volume.

Importance of data reduction:

⇒ Huge datasets can lead to impractical or infeasible analysis.

⇒ Data reduction techniques offer a solution to obtain a smaller yet representative dataset.

Characteristics of reduced data:

⇒ maintain the integrity of data.

⇒ Much smaller in volume.

⇒ By reduction efficiency of the data mining process is improved.

⇒ Data reduction does not affect the result obtained from data

mining.
⇒ That mean the result obtained from data mining before and after data reduction is same or almost the same.

Why data reduction?

⇒ A database/ data warehouse may store terabytes of data.
⇒ Complex data analysis/ mining may take a very long time to run on the complete data set.
⇒ Large dataset size poses challenge for complex analysis and mining.

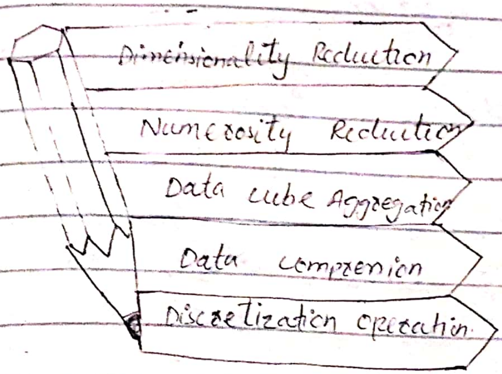
Three operations in data reduction process

⇒ The three basic operations in a data reduction process are:
1. Delete a column
2. Delete a row
3. Reduce the numbers of values in a column (smooth a feature).

⇒ These operations attempt to preserve the character of the original data by deleting data that are non-essential.

Techniques of data Reduction

⇒ These are the following techniques or methods data reduction in data mining that are:



1. Dimensionality reduction:

⇒ The number of input features, variables dataset is known as dimensionality and the process to reduce these features is called dimensionality reduction.
⇒ It is a way of converting the higher dimensions dataset into lesser dimensions dataset ensuring that it provides similar information.
⇒ it reduces or eliminates outdated or redundant features.

⇒ Methods of dimensionality reduction:

⇒ Wavelet Transform

⇒ feature reduction ⇒ { Feature selection / Feature extraction }

⇒ Relief Algorithm

⇒ PCA

⇒ Attribute-subset selection

Definition:

⇒ Process of reducing the number of random variables or attributes under consideration.

I. Attribute-subset selection:

⇒ method of dimensionality reduction in which irrelevant, weakly relevant, or redundant attributes or dimensions are detected and removed.

⇒ Also known as feature subset selection.

⇒ Basic heuristic methods of attribute subset selection include the following techniques:

1. Stepwise forward selection:

- The procedure starts with an empty set of attributes as the reduced set.
- The best of the original attributes is determined and added to the reduced set.
- At each subsequent iteration or step, the best of the remaining original attributes is added to the set.

Ex:

Initial attribute set:

$\{A_1, A_2, A_3, A_4, A_5, A_6\}$

Initial reduced set:

⇒ { }

⇒ $\{A_1\}$

⇒ $\{A_1, A_4\}$

⇒ Reduced attribute set:

$\{A_1, A_4, A_6\}$

2. Stepwise backward elimination:

- The procedure starts with the full set of attributes.
- At each step, it removes the worst attribute remaining in the set.

Ex:

Initial attribute set:

$\{A_1, A_2, A_3, A_4, A_5, A_6\}$

$\Rightarrow \{A_1, A_3, A_4, A_5, A_6\}$

$\Rightarrow \{A_1, A_4, A_5, A_6\}$

$\Rightarrow$ Reduced attribute set:

$\{A_1, A_4, A_6\}$

3. Combination of forward selection and backward selection:

- The stepwise forward and backward elimination methods can be combined so that, at each step, the procedure selects the best attribute and removes the worst from among the remaining attributes.

4. Decision tree induction:

⇒ Decision tree algorithms (ID3, C4.5 & CART) were originally intended for classification.

⇒ Decision tree induction constructs a flowchart like structure where each internal (nonleaf) node denotes a test on attribute

⇒ Each branch corresponds to an outcome of the test

⇒ Each external (leaf) node denotes a class prediction.

⇒ At each node, the algorithm chooses the "best" attribute to partition the data into individual classes.

⇒ When decision tree induction is used for attribute subset selection, a tree is constructed from the given data.

⇒ All attributes that do not appear in the tree are assumed to be irrelevant.

⇒ The set of attributes appearing in the tree form the reduced subset of attributes.

Attribute construction:

⇒ Sometimes we may create new attributes based on others.

⇒ Such attribute construction can help improve accuracy and understanding of structure in high dimensional data.

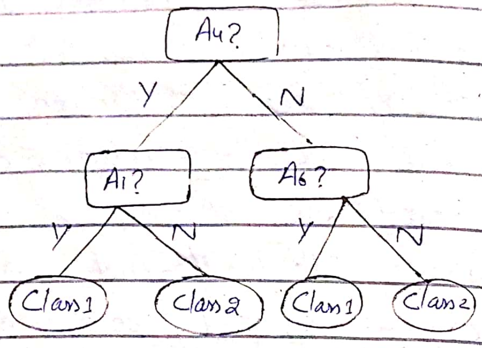
for example:

- Create an attribute area by combining height and width.

Decision Tree induction

Initial attribute set:

{A1, A2, A3, A4, A5, A6}



⇒ Reduced attribute set:

{A1, A4, A6}

Note:

⇒ Attribute subset-selection method also known as feature selection.

Feature selection:

- It is the process of selecting the subset of relevant features and leaving the irrelevant features present in the dataset to build a model of high accuracy.

⇒ feature selection also known as variable selection, feature reduction, attribute selection, or variable subset selection.

Objective of feature selection:

The objective of feature selection:

- Improving the performance of data mining model
- Providing a faster and more cost-effective learning process
- Providing a better understanding of the underlying process that generates the data.

Categories of feature selection algorithms:

⇒ Fall into two categories:

1. Feature ranking
2. Subset selection.

⇒ Feature-selection methods are applied in one of three conceptual frameworks:

- Filter model (information gain)
- wrapper model (Accuracy)
- Embedded methods. (Errors, add or remove)

Filter model: → Based on the information theory.
→ Assigns the rank to each feature

⇒ Features are selected as a preprocessing step.

=> Selection is based on an evaluation function using search methods.

=> Typically involves heuristic search due to the impractically of exhaustive search.

=> Evaluation is independent of specific data mining techniques.

Wrapper models:
→ Defines the subset of feature set
→ use the searching algorithm.

=> Features are selected by wrapping the search around the chosen learning algorithm.

=> Feature subsets are evaluated based on the performance of the learning algorithm.

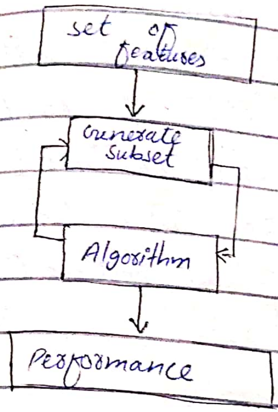
=> Computational complexity is a drawback due to evaluating multiple feature subsets.

Embedded methods:

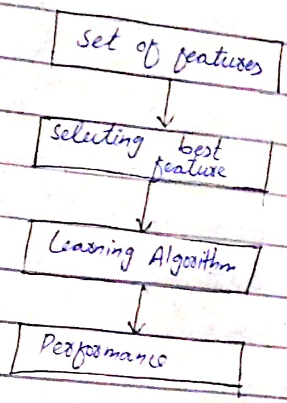
- Feature selection and learning algorithm are integrated into a single optimization problem.
- Efficient for large datasets with many samples and dimensions.

=> To sum up, in feature selection finding k of the total of n features that give us the most information and discard the others (n-k) dimensions.

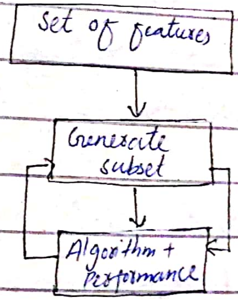
Wrapper method.



Filter method



Embedded method



⇒ Process of transforming the space containing many dimensions into space with fewer dimensions. Useful when we want to keep the whole information but use fewer resources while processing the information.

II. Feature Extraction:

⇒ Transforming the input set into a new, reduced set of features is called feature extraction.

⇒ If the features extracted are carefully chosen, it is expected that the new feature set will extract the relevant information from the input data in order to perform the desired data-mining task using this reduced representation.

Aim:

⇒ Feature transformation techniques aim to reduce the dimensionality of data to a small number of dimensions that are linear or non-linear combinations of the original dimensions.

⇒ Common and widely used method is PCA and Linear Discriminant analysis LDA.

Why feature extraction?

1. To improve the performance of the model generation process and the resulting model itself.
2. To reduce dimensionality of the model without reduction of its quality through:
   (a) elimination of irrelevant features
   (b) Detection and elimination of redundant data and features.
   (c) Identification of highly correlated features.
   (d) extraction of independent features that determine the model.
3. To help the user visualize alternative results, which have fewer dimensions, to improve decision making.

Benefits:

- Improved computational efficiency
- Enhanced interpretability
- Better performance.

III: Relief Algorithm

⇒ A releief algorithm is a feature selection technique used to identify and select the most relevant features from a dataset while removing redundant or irrelevant ones.

⇒ Feature weight-based algorithm for feature selection inspired by instance-based learning.

⇒ Relief algorithm work by evaluating relevance of each feature in a labeled training dataset for classification problems.

⇒ main ideas

- Compute a ranking score for each feature indicating its ability to separate neighbouring samples.

⇒ The author of the Relief algorithm Kira and Rendell, proved that the ranking score is large for relevant features and small for irrelevant ones.

Algorithm steps

Initialization:

1. Randomly select a subset of samples of size m from the training data set. where m is a user defined parameter.

For each randomly selected sample x:-

2. For each randomly selected sample X from a training data set, Relief searches for its two nearest neighbours:
   - One from the same class, called nearest hit H.
   - Other one from a different class, called nearest miss M.

3. Update the quality score for each feature based on the differences in their values for the sample, nearest hit and nearest miss.

$$W_{new}(A_i) = W_{old}(A_i) - (diff(X[A_i], H[A_i])^2 + diff(X[A_i], M[A_i]^2)/m$$

4. Repeat steps 1-3 for a total of m times.

Thresholding:-

5. Accumulate the scores for each feature and determine those features with scores greater than or equal to a predefined threshold T as statistically relevant to the classification task.

Initialize: $W(A_j) = 0$; $i = 1, \ldots, P$ (P is the number of features)

For $i = 1$ to m

i) Randomly select sample X from training data set S.

Find nearest hit H and nearest miss M samples.

For $j = 1$ to P

$W(A_j) = W(A_j) - (diff(X[A_j], H[A_j])^2 + diff(X[A_j], M[A_j]^2)/m$

End.

End.

Output: Subset of feature where $W(A_j) \geq \tau$

⇒ Relief is a simple algorithm that relies entirely on a statistically methodology.

⇒ The algorithm employs few heuristics and therefore it is efficient.

⇒ Scalability and efficiency:

- Relief algorithm has computational complexity $O(Pn)$, making it scalable to datasets with large sample size(n) and high dimensionality (P).
- It is one of the few algorithms that can evaluate features for real-world problems with large feature space and large number of samples.
- It's noise tolerant and unaffected by feature interaction, suitable for hard data-mining algorithms.

Limitations:

- Does not remove redundant features: relevant features may still be highly correlated.
- Proper value of threshold τ is crucial, typically determined through experimentation.

Extensions

- Relief algorithm has been extended to handle multi-class problems, noise, redundant and missing data.
- Variants like ReliefF, Simba, and I-Relief have been proposed as improvements.

Example:

⇒ Given a training dataset with three features (F_1, F_2 and target class) and four samples.

⇒ Relief calculates scoring values for features F_1 and F_2 based on their differences in values for each sample, nearest hit, and nearest miss.

Training data set for applying Relief Algorithm:

Sample	F_1	F_2	Class
1	3	4	C_1
2	2	5	C_1
3	6	7	C_2
4	5	6	C_2

$W(F_1) = (0 + [-1+4] + [-1+9] + [-1+9] + [-1+4]) / 4 = 5.5$

- ~~$W(F_1) = (0+1-1+4-1+1-4+1-4)/4$~~ $= 5/4 = 1.25$
- $W(F_2) = (0 + [-1+4] + 1 + [[-1+4] + (1+1)) / 4 = 5/4 = 1.25$

⇒ These scoring values indicate the relevance of features F_1 and F_2 in distinguishing between samples from different classes in the dataset.

ReliefF:

⇒ The ReliefF algorithm is an extension of the relief algorithm.

⇒ It is smarter version of relief.

⇒ It is designed to work better with datasets where some categories have a lot or more examples than others.

⇒ Specifically designed to handle features selection tasks in datasets with imbalanced class distribution.

⇒ ReliefF looks at how different features are between similar and different examples to decide which ones are most important for making predictions.

⇒ This helps it pick out the right features even when the data is unevenly spread out across different categories.

SURF:

- Stands for Speeded up Robust features.
- It is a feature detection and description algorithm used in computer vision for tasks like object recognition and image stitching.
- SURF finds special points in images called keypoints that are unique and stay consistent even if the image changes.
- It figures out the size and angle of each keypoint to make sure it stays useful even if the image is rotated or resized.
- It is designed to be fast, so it can work quickly even on large images.
- Used in robotics, virtual reality, image stitching.
- well suited for real-time applications.

To sum up:-

- The relief algorithm iteratively evaluates the relevance of each feature by comparing its values between samples and their nearest hits and misses, updating quality scores accordingly.
- It then identifies features that are statistically relevant for classification based on a user-defined threshold.

⇒ Classes refer to the categories or labels assigned to the samples in a classification problem.

⇒ Each sample in the dataset belongs to a specific class, which represents the category or group it belongs to.

example:

- For example in binary classification problem where we are trying to predict whether an email is "spam" and "not spam".
- In multi-class classification problem, such as classifying images of animals into categories like "dog", "cat", "bird" and "fish"; each of these categories represents a class.

Wavelet Transform:

⇒ The discrete wavelet transform is a linear signal processing technique that allowing to represent data in a numerically different form while retaining important information.

⇒ By transforming a data vector A into a new vector A', both of the same length, we create opportunities to reduce data size effectively.

⇒ Wavelet transform enables data reduction by truncating or discarding less significant information.

⇒ After transformation we retain only the most important wavelet coefficients, leading to compressed data.

⇒ This process involves keeping the smallest fragment of the strongest wavelet coefficients, thus achieving compression while maintaining essential data features intact.

⇒ Wavelet transform is versatile and applicable to various types of data including data cubes, sparse data, or skewed data.

PCA:-

introduced by mathematician Karl Pearson in 1906

⇒ PCA stands for Principal Component Analysis.

- is an unsupervised learning algorithm technique - determine interrelation among set of variables

⇒ PCA, also known as the Karhunen-Loeve (K-L) method.

- Also known as general factory analysis

⇒ Widely used technique for dimensionality reduction.

- Uses an orthogonal transformation that converts a set of correlated variables to a set of uncorrelated variables.

⇒ It finds important patterns in the data or principal components and reduces its complexity.

⇒ It searches for orthogonal vector to represent data effectively.

Computation of principal components

- PCA computes k orthonormal vectors serving as the basis for the normalized input data.
- These vectors, termed principal components, are perpendicular to each other.
- The number of PC needed for good representation of the data is determined by analyzing the proportion of variance.

⇒ Principal Components are computed as the eigenvectors of the covariance matrix of the features.
⇒ These components define a linear transformation from the original n-dimensional space to m-dimensional space.
⇒ It is an un-supervised learning algorithm.
⇒ It is a statistical process that converts the observations of correlated features into a set of linear un-correlated features with the help of orthogonal transformation.

PCA - Algorithm :-

Step 1: Data

- Consider a dataset having n features or variables denoted by $X_1, X_2, \ldots, X_n$.
- Let there be N examples.
- Let the value of the i^{th} feature X_i be $X_{i1}, X_{i2}, \ldots, X_{iN}$.

Feature	Example 1	Example 2	Example N
X_1	X_{11}	X_{12}	X_{1N}
X_2	X_{21}	X_{22}	X_{2N}
$\vdots$ X_i	X_{i1}	X_{i2}	X_{iN}
$\vdots$ X_n	X_{n1}	X_{n2}	X_{nN}

Step 2:

- Compute the means of the variables.

$$\bar{X}_i = \frac{1}{N}(X_{i1} + X_{i2} + \cdots + X_{iN})$$

Step 3:

- Calculate the Covariance matrix.

$$Cov(X_i, X_j) = \frac{1}{N-1} \sum_{k=1}^{N} (X_{ik} - \bar{X}_i)(X_{jk} - \bar{X}_j)$$

$$S = \begin{bmatrix} Cov(X_1, X_1) & Cov(X_1, X_2) & \cdots & Cov(X_1, X_n) \\ Cov(X_2, X_1) & Cov(X_2, X_2) & & Cov(X_2, X_n) \\ \vdots & & & \\ Cov(X_n, X_1) & Cov(X_n, X_2) & \cdots & Cov(X_n, X_n) \end{bmatrix}$$

Step 4:

Calculate the eigenvalues and eigenvectors of the covariance matrix.

i) Set up the equation:

- This is a polynomial equation of degree n in λ. It has n real roots and these roots are the eigenvalues of S.

$$\det(S - \lambda I) = 0$$

ii) if $\lambda = \lambda'$ is an eigenvalue, then the corresponding eigen vector is a vector

$$U = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad \text{such that } (S - \lambda' I)U = 0$$

iii) Now, normalize the eigenvectors:

- Given any vector X normalize it by dividing X by its length.
- The length (or the norm) of the vector

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

is defined as

$$\|X\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

- Compute the n normalized eigenvectors $e_1, e_2, \dots, e_n$ by

$$e_i = \frac{1}{\|U_i\|} U_i, \quad i = 1, 2, \dots, n$$

Steps:

- Derive new data set.
- Order the eigenvalues from highest to lowest.
- The unit eigenvector corresponding to the largest eigenvalue in the first principal component.

i) Let the eigenvalues in descending order be $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ and let the corresponding unit eigenvector $e_1, e_2, \dots, e_n$.

ii) Choose a positive integer P such that $1 \leq P \leq n$.

iii) Choose the eigenvectors corresponding to the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_P$ and form the following $P \times n$ matrix.

iv) Form the following $n \times N$ matrix:

$$X = \begin{bmatrix} X_{11} - \bar{X}_1 & X_{12} - \bar{X}_1 & \cdots & X_{1N} - \bar{X}_1 \\ X_{21} - \bar{X}_2 & X_{22} - \bar{X}_2 & \cdots & X_{2N} - \bar{X}_2 \\ \vdots & & & \\ X_{n1} - \bar{X}_n & X_{n2} - \bar{X}_n & \cdots & X_{nN} - \bar{X}_n \end{bmatrix}$$

v) Next compute the matrix:

$$X_{new} = FX.$$

This is $P \times N$ matrix. gives us dataset of N samples having P features.

$$P_{ij} = e_i^T \begin{bmatrix} X_{1j} - \bar{x} \\ X_{2j} - \bar{x}_2 \\ \vdots \\ x_{nj} - \bar{x}_n \end{bmatrix}$$

Application:

- PCA can handle ordered and unordered attributes, sparse data, and skewed data.
- It reduces multi-dimensional data to two dimensions for analysis.
- PC can serve as i/p for multiple regression and cluster analysis.

Comparison with Wavelet Transforms:

- PCA is adapt at handling sparse data.
- Wavelet transforms excel with high-dimensional data.

Why is PCA useful?

- It helps in understanding large datasets by reducing their size and Complexity.
- PCA reveals hidden patterns that can be valuable for decision making.

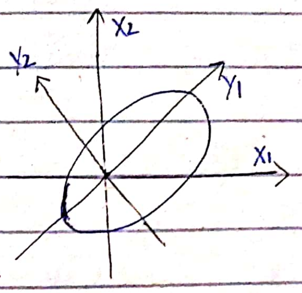
Real Life Example:

- Imagine you have a lot of data about different flowers.
- PCA can help find the most important features, like petal size and color, to understand and compare them more easily.

Another name:

⇒ In various fields, it is also known as the SVD, the Hotelling transform and the empirical orthogonal function (EOF) method.

⇒ PCA is a method of transforming the initial data set represented by vector samples into a new set of vector samples with desired dimensions.



⇒ Y_1 & Y_2 are the first two PC for the given data.

2. Numerosity Reduction

⇒ Numerosity refers to the quantity or number of elements in a set or collection.
⇒ Numerosity reduction techniques aim to reduce the size or volume of data while preserving essential information or patterns.
⇒ It can lead to efficient storage, processing, and analysis of data, making it easier to handle large datasets and extract meaningful insights.

Types of Numerosity Reduction

⇒ There are two Types of numerosity reduction:

1. Parametric
2. Non- Parametric

i) Parametric:

Definition:

⇒ Parametric numerosity reduction involves storing only data parameters instead of the original data.

Method of Parametric Numerosity:

1. Linear Regression:

⇒ Models a relationship between two attributes using a linear equation.
⇒ Equation: $y = wx + b$

⇒ Here, y is the response attribute
x is the predictor attribute and
w and b are regression coefficients.
⇒ It is like drawing a straight line through a scatter plot of points.
⇒ The method of least squares is commonly employed to determine the coefficients.
⇒ Helps understand how one thing changes with another.
⇒ Multiple linear regression extends this to model linear functions between two or more predictor variables.

2. Log Linear model:

⇒ Estimates probabilities in multidimensional data.
⇒ Discovers relationships between two or more discrete attributes in the database.
⇒ Useful for simplifying complex data into manageable patterns.
⇒ This approach facilitates dimensionality reduction and data smoothing, as lower-dimensional points occupy less space and are less prone to sampling variations.

Application & Scalability:

⇒ Both methods work for sparse and skewed data.
⇒ Linear regression is great for skewed data, while log-linear models handle high dimensional data better.
⇒ Linear regression might get slow with lots of dimensions, but log linear handle it well up to around 10 dimensions.

⇒ There are tools like SAS, SPSS and S-Plus that help with regression.

(ii) Non-parametric:

⇒ Does not relay on any specific model assumptions.

⇒ Leads to more uniform reduction, regardless of data size.

⇒ May not achieve as a high level of data reduction as parametric techniques.

Methods of Non-parametric

⇒ There are atleast four types of Non-parametric data reduction techniques:

- Histogram
- Clustering
- Sampling
- Data Cube Aggregation
- Data Compression

1. Histogram

Definition:

⇒ A graphical representation of frequency distribution of data.

⇒ Frequency distribution describe how often a value appear in a data set.

⇒ Histograms use binning to approximate data distributions.

⇒ They help in reducing the complexity of data by organizing it into discrete subsets of bins.

Partitioning data with Histograms

Bucket Determination:

⇒ Histograms partition the data distribution into disjoint subsets or bins, known as buckets.

⇒ Bucket can represent either single attribute value / frequency pairs or continuous ranges for the attribute.

Partitioning Rules

Equal-width: Each bucket has a uniform width.

Equal frequency (or equal depth):

- Buckets are created so that each contains roughly the same number of data samples.

Effectiveness of Histograms

⇒ Histograms are highly effective in approximating various types of data:
- They work well with sparse and dense data.
- They handle highly skewed and uniform data distribution effectively.

Extension to multiple Attributes

⇒ Single attribute histograms can be extended to handle multiple attributes.

Singleton Buckets:

⇒ Singleton buckets are useful for storing high-frequency outliers.

2. Sampling:

Definition:

⇒ Sampling involves selecting a smaller random subset from a large dataset to represent it more efficiently.

Types of Sampling

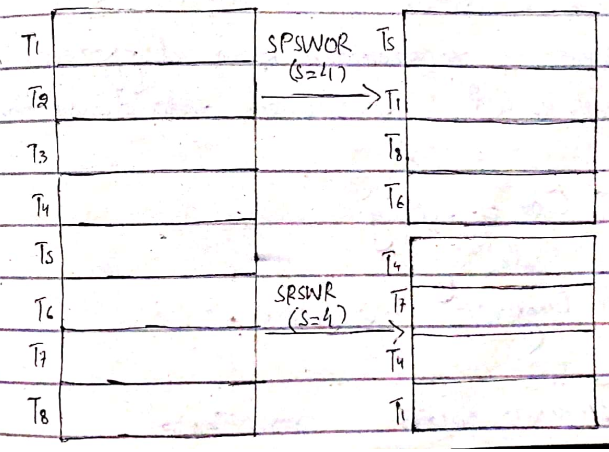
1. Simple random Sample without Replacement:

⇒ Involves randomly selecting a subset of size s from the dataset without replacing the selected tuples.

⇒ Each tuple has an equal chance of being selected. 1/N.

2. Simple random Sample with replacement:

⇒ Similar to SRSWOR but allows selected tuples to be replaced back into the dataset for potential re-selection.



3. Cluster Sample:

=> Involves grouping tuples into mutually disjoint clusters.

=> A subset of clusters (of size s) is randomly selected.

=> Useful for databases where tuples are retrieved in clusters, such as by pages.

ex:

In previous example cluster sample (s=2)

4. Stratified Sample:

=> Divides the dataset into mutually exclusive parts called strata.

=> A random subset (of size s) is selected from each stratum.

=> Ensures representation from different groups, useful for skewed data distribution.

Advantages of Sampling:

- Cost efficiency
- Scalability
- Progressive Refinement.

Applications:

1. Aggregate queries 2. Error estimation

3. Clustering:

Definition:

=> Clustering techniques organize similar objects from the data into groups called clusters.

=> Groups are made based on the similarity of data tuples to one another and ~~dissensi~~ dissimilarity to objects in other clusters.

Object Representation:

=> Data tuples are treated as objects to be grouped together based on similarity.

Cluster Quality:

=> Diameter: Represents the maximum distance between any two objects within a cluster.

=> Centroid Distance: Measures the average distance of each object from the cluster centroid.

Similarity Measures:

=> Typically defined using a distance function, determining how close objects are in space.

Effectiveness:

=> Effectiveness of clustering depends on the nature of the data:

- Effective for data organized into distinct clusters.
- Less effective for data with smeared or dispersed patterns.

Measure for defining Clusters:

=> Various measures exist for defining clusters such as:

- Density-based clustering
- Connectivity-based
- Distribution-based
- Centroid-based
- Hierarchical clustering

Ex:

Customer data in a City.

4. Data Cube Aggregation:

=> A technique used to aggregate data into simple form.

=> Aggregates data at various levels of data cube to represent the original dataset.

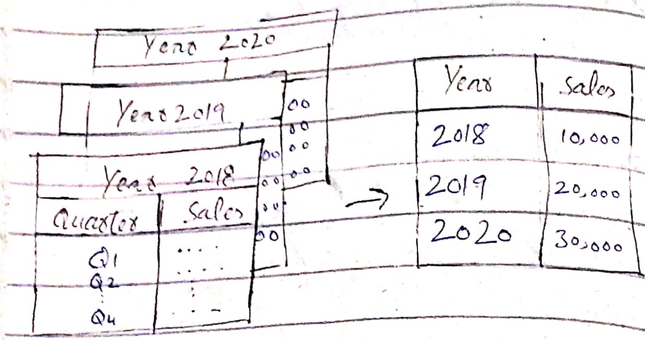
Process:

=> Aggregate data at different levels of granularity within a data cube.

=> Summarizes and precomputes data to provide faster access and analysis.

Example:

- Consider Sales data for All Electronics from 2018 to 2022, recorded quarterly.
- To obtain annual sales per year, aggregate the sales data for each quarter within each year.
- This provides ~~an~~ the required data in much smaller size without losing any information.



5. Data Compression

Definition:

⇒ Modifying or encoding, or restruction data to occupy less space.

Objective:

⇒ Build a compact representation of information by removing redundancy and representing data in binary form.

Types of data Compression

1: Lossless:

⇒ Allows restoration of the exact original data from compressed data.

Techniques:

- Run-Length encoding

Usage:

- Ideal for scenarios where preserving every detail of the original data is critical.

2. Lossy Compression:

⇒ Result in some of loss data during Compression.

Decompressed data:

May differ slightly from the original, but still useful for retrieving information.

Examples:

JPEG image format employs lossy compression.

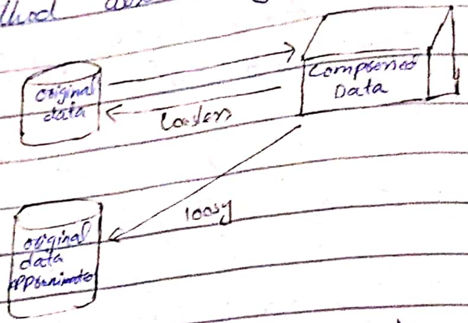
Techniques:

- Discrete Wavelet Transform
- PCA

Compression techniques:

- Huffman Encoding
- Run-Length encoding

Note:
Dimensionality and numerosity reduction method also used for data compression.



5. Discretization Operation

⇒ Divide the attributes of the continuous nature into data with intervals.

Types

1. Top-down
Consider one point called breakpoints or split points to divide the whole set of attributes and repeat this method up to end, called top-down, also known as splitting.

2. Bottom-up
o By considering all the constant values as split points, some of discarded through a combination of neighbourhood values in the interval, called bottom up.

Q. Write down steps of ChiMerge algorithm for discretization?

⇒ ChiMerge algorithm consists of three basic steps for discretization.

1. Sort all the data for the given feature in ascending order.
2. Define initial intervals so that every value of the feature is in a separate interval.
3. Repeat until no χ^2 of any two adjacent intervals is less then threshold value.

$$\chi^2 = \sum_{i=1}^{2} \sum_{j=1}^{k} (A_{ij} - E_{ij})^2 / E_{ij}$$

Q. Write down characteristics of data reduction algorithms?

⇒ Data reduction algorithms are characterized by several key characteristics.

1. Measurable Quality:
⇒ Easily measure how good the approximate results are.
⇒ The effectiveness of a data reduction algorithms can be quantitatively measured.

2. Recognizable quality:-

⇒ The quality of the reduced data can be assessed by humans.

⇒ Easily determine the quality of the approximated results while the algorithm is running, even before applying an data-mining procedures.

3. Monotonicity:-

⇒ These algorithms usually work in steps, and the quality of results keeps getting better as it goes on.

⇒ It's like climbing stairs - you keep going up without going down.

⇒ The results keep getting better over time without getting worse.

4. Consistency:-

⇒ The quality of results is directly related to how much time it takes to compute and the quality of the i/p data.

⇒ This means that if you run the algorithm multiple times on the same data set, you should get the same o/p each time.

5. Diminishing Returns:-

⇒ At the beginning of the computation, the improvements in the solution are significant.

⇒ Improvements in the solution become smaller as the algorithm progresses.

6. Interruptability:-

⇒ The algorithm can be stopped at any time and still get some useful results.

7. Preemptability:-

⇒ The algorithm can be pause/suspended and resumed with minimal overhead or extra work.

Q. Write down feature-reduction process results?

⇒ A feature-reduction process should result in:

1. Less data so that the data-mining algorithm can learn faster.
2. Higher accuracy of a data-mining process so that the model can generalize better from the data.
3. Simple results of the data-mining process so that they are easier to

understand and use.

4. Focus features, so that in the next round of data collection, savings can be made by removing redundant or irrelevant features.

=> The purpose of feature-reduction process vary, depending upon the problem in hand and for specific application, but generally, we want:

1. To improve performances
2. To reduce dimensionality of the model without reduction of its quality through:
   a) elimination of irrelevant features
   b) Detection and elimination of redundant data and features
   c) Identification of highly correlated features.
   d) Extraction of independent features that determine the model.
3. To help the user visualize alternative results, which have fewer dimensions, to improve decision making.

Q. Write down two basic forms of random sampling?

A) The two basic forms of random sampling are as follows:

1. Incremental Sampling:

what it does?

=> It looks at random parts of the data, starting small and gradually getting bigger.

Why: To see how things change as we look at more data.

How it works:

=> Start with a small chunk of data, like 10%, then look at 20%, 33%, and so on, up to 100%.

When to stop:

=> When don't see any more changes or improvements.

2. Average sampling:

what it does?

=> Combines solutions from different random parts of the data.

Why?
=> To get a better overall result.
How it works?
=> Look at different parts of the data and find solutions.
=> Then average them together.
For what?
=> For classification we vote.
=> For regression we average.
when to use?
=> when we want to make sure our final result is reliable and accurate.

Q. Define Stratified Sampling?

=> It divides the whole dataset into smaller non-overlapping groups called strata.
=> Samples are taken independently from each stratum.
=> Why use it?
=> when the groups are similar within themselves but different from each other.
=> Want a more accurate estimate with less variability.

Example:
=> if we are studying student performance, divide them into strata based on grade levels e.g (9th, 10th, 11th, 12th), then sample from each grade level independently.

Q. Define inverse sampling?

=> It is used when something happens very rarely in the data.
=> Start with a small sample and keep adding more until have enough information about the rare event.
Why use?
=> When even a large sample may not give enough information about the rare event.
Example:
=> if we are studying rare diseases, we might start with a small sample of patients and keep adding more until we have enough cases to study.
=> These techniques help make sampling more efficient and accurate for different types of data.